Lecture (1)

INTRODUCTION TO COMPUTER SKILLS FOR MEDICAL STUDENTS

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Outlines

- Introduction
- Course outlines
- Assessments
- Requirements

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Course Outlines

Theory part:

- Computers in medical fields and medical education
- Types and Components of computer.(Software and Hardware)
- Introduction to Operating system.
- Introduction to Internet and Internet services.
- Preparing medical documents.
- Preparing medical presentations.
- Preparing medical calculations and statistical sheets.

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Practical Part:

- windows 10 environment.
- Searching techniques.
- Introduction to Microsoft office software.
- Microsoft office word 2013.
- Microsoft office power point 2013.
- Internet Cloud Services (Google and outlook services).
- Google Form in medical research.
- Microsoft office Excel 2013.

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Assessments

-
- Projects 30%
- Class work & Tests 10%
- Final exam 60%

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Objectives Overview

Explain why computer literacy is vital to success in today's world

Define the term, computer, and describe the relationship between data and information

Role of computer in Medical Fields

Role of Computers in Medical Education

Discuss the advantages and disadvantages that users experience when working with computers

See Page 3
for Detailed Objectives

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A World of Computers

- Computers are everywhere

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What Is a Computer?

- A Computer is a device or set of devices that work under the control of a stored program, automatically accepts and processes data to provide information.
- Computers are become integral elements in public health and biomedical research, medical education, patient care and enhanced information management.

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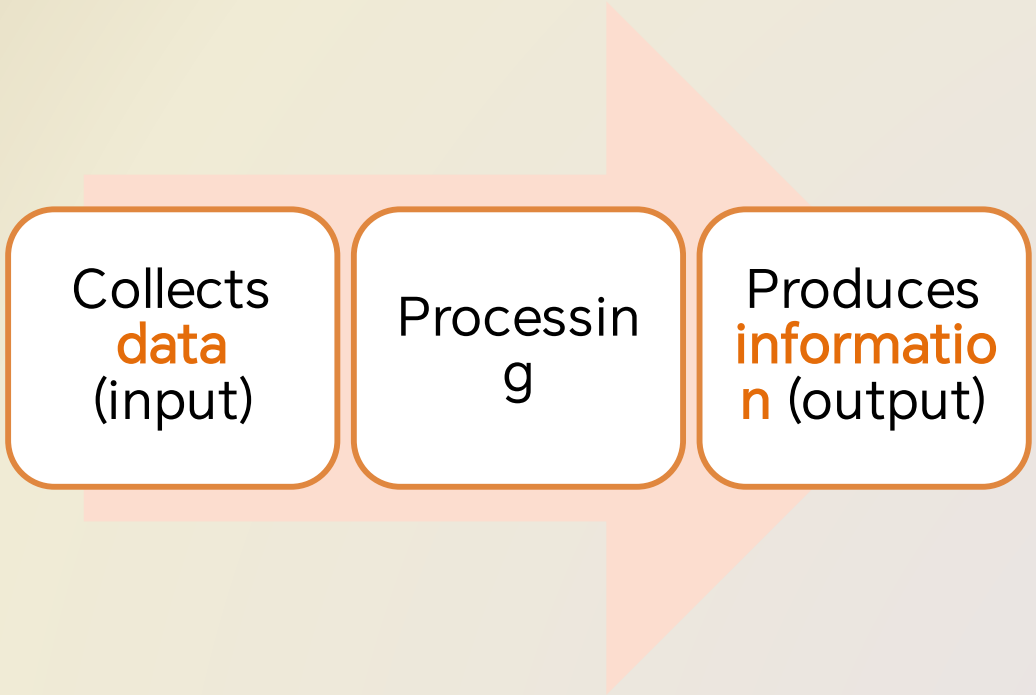
What Is a Computer?

Information Processing Cycle



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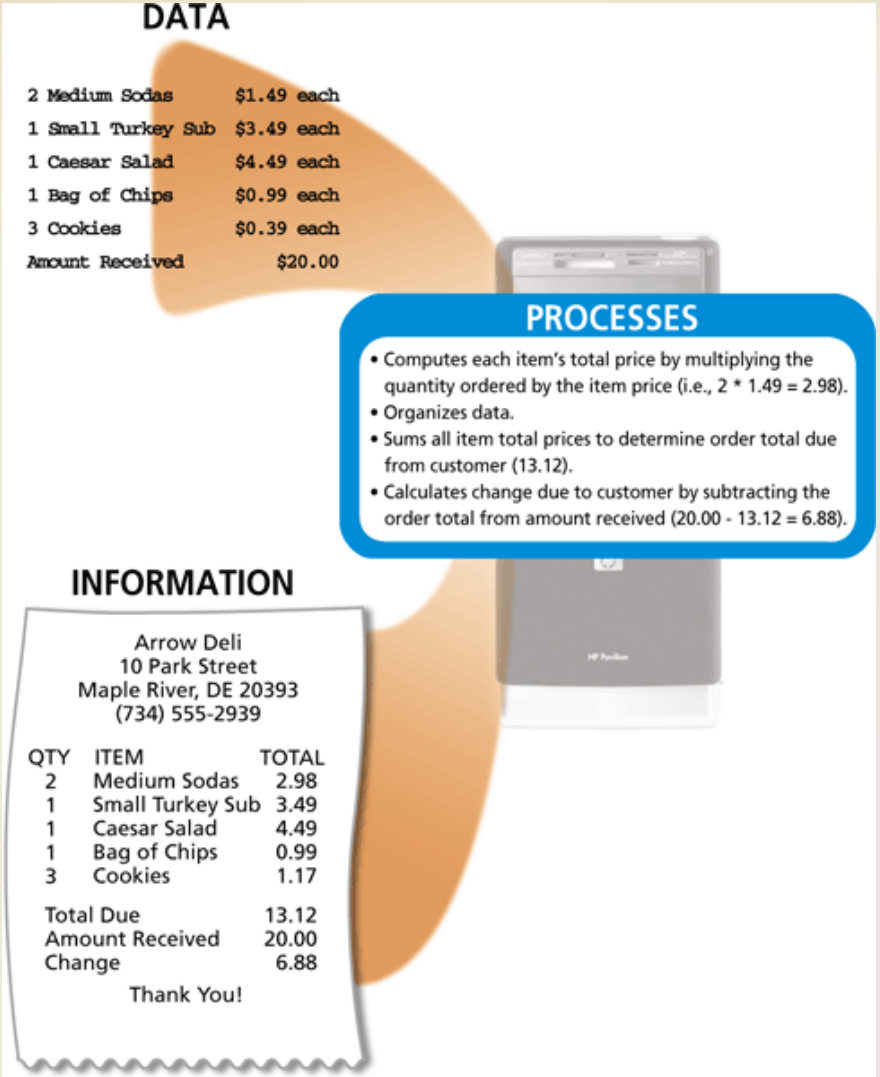

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Role of computer in Medical Fields

- Computers play a key role in almost every sphere of our life. Computers can hence be employed in a wide variety of fields like engineering, education as well as health and medicine.

Role of computer in medical fields:

- Computers are the excellent means for storage of patient related data. Computers enable an efficient storage of huge amounts of medical data.

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Cont...

- Medical information can be effectively stored in a computer database (patient's family history, physical ailments, already diagnosed diseases and prescribed medicines.)
- Computers can keep track of prescriptions and billing information.

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Cont...

- Modern medical equipments have small programmed computers and instructions such as:
  - Hospital-bed beeping systems
  - Emergency alarm systems
  - X-ray machines.

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Cont…

- Computer software can be used for:
  - diagnosis of diseases
  - examination of internal organs of the body.
- Computer scanning used for medical imaging to create images of the human body.
- Computers allow faster communication between a patient and a doctor, and better collaboration between doctors over the Internet

Role of Computers in Medical Education

What is a computer in Medical education?

- It means using Computers as learning methods to allowing students to explore a body of materials using computer methods which are best suited to their own learning styles.

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Cont...

1) computer can provide quick access to reference and new content.
2) Computer can supplement the static images with sounds, video clips, and interactive teaching modules. Immersive interfaces, which present three-dimensional worlds

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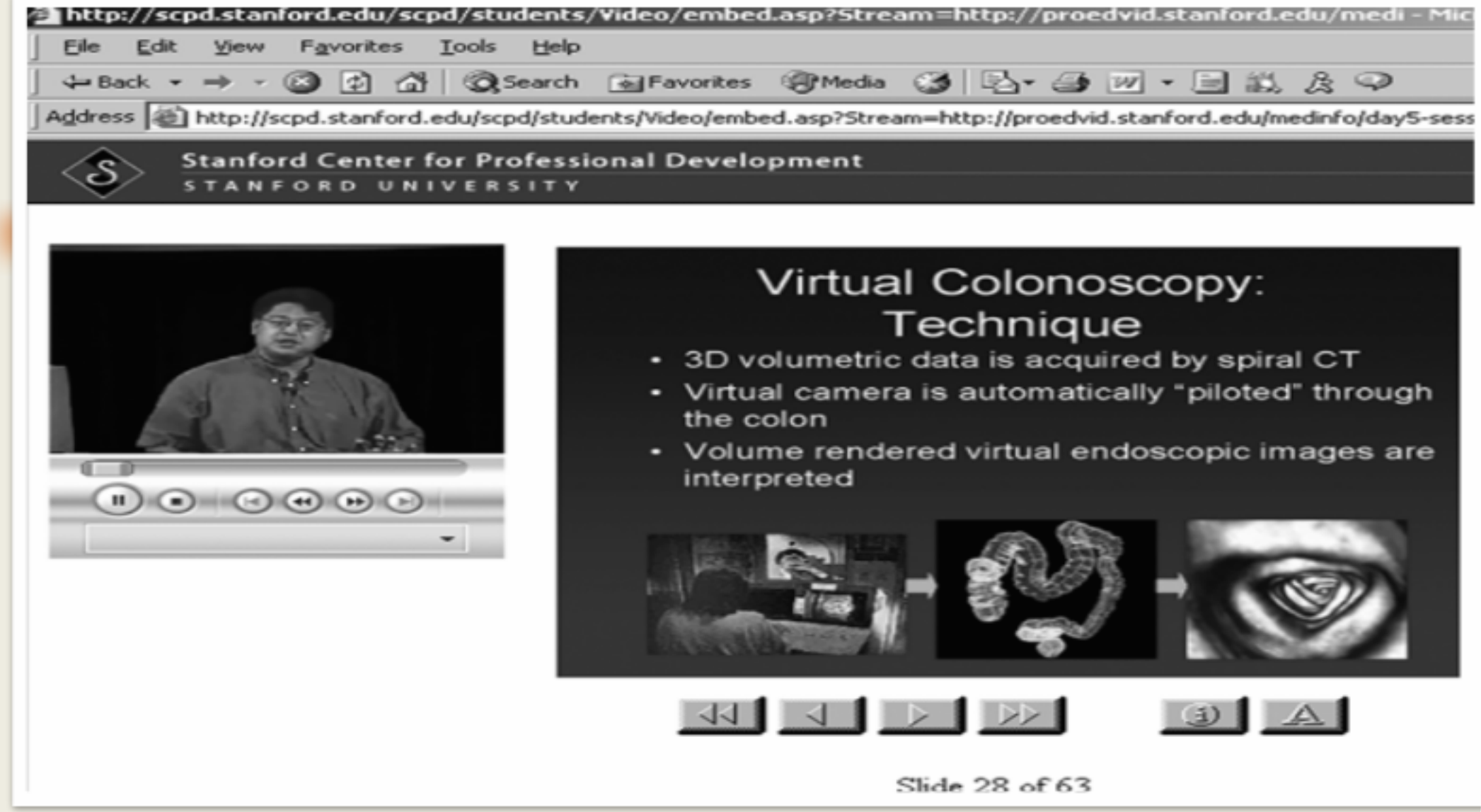

Figure 1.2 A digital video lecture is presented within a browser for the Web.
The video image in the upper left is augmented with high-resolution images of the lecture slides on the right.

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Cont…

3) Computer-based learning can take place at the time and location best suited to the needs of the learner:

"Any time, any place" learning becomes practical.

4) computer-based learning can be enjoyable and engaging, maintaining the interest of the student.

5) Computer storage can serve housing the medical information in an electronic format:

- medical journals
- research and diagnosis papers
- important medical documents
- reference books

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Cont...

6) Deliver teaching materials, with the advantage of the removal of time and space limitations.(recorded lectures , digitized videos of the lectures as well as the related slides) .

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Cont...

7) Multi-choices Exam system

- Drill and practice was the first widespread use of computer-based learning, developed almost as soon as computers became available.
- The computer grades the selected answers and, based on the accuracy of the response, repeats the teaching material, or allows the student to progress to new material.

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Advantages and Disadvantages of Using Computers

Advantages of Using Computers

- Speed
- Reliability
- Consistency
- Storage
- Communications

Disadvantages of Using Computers

- Health Risks
- Violation of Privacy
- Public Safety
- Impact on Labor Force
- Impact on Environment

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END
QUESTION ????